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COMMON HALF YEARLY EXAMINATION - 2025	Reg. No.						

XII - MATHEMATICS

Time Allowed : 3-00 Hrs. *Maximum Marks: 90*

Part – I

I. Choose the correct answer: **20 x 1 = 20**

1. $A^T A^{-1}$ is symmetric, then A^2 =
   a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$
2. The area of the triangle formed by complex numbers z, iz and z + iz in the Argand's diagram is
   a) $\frac{1}{2}|z|^2$ b) $|z|^2$ c) $\frac{3}{2}|z|^2$ d) $2|z|^2$
3. If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum\frac{1}{\alpha}$ is
   a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$
4. If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to
   a) $[-1,1]$ b) $[\sqrt{2}, 2]$ c) $[-2,\sqrt{2}]\cup[\sqrt{2},2]$ d) $[-2,-\sqrt{2}]$
5. The area of the quadrilateral formed with foci of the hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ is
   a) $4(a^2 + b^2)$ b) $2(a^2 + b^2)$ c) $(a^2 + b^2)$ d) $\frac{1}{2}(a^2 + b^2)$
6. If the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$, then (α, β) is
   a) (–5,5) b) (–6,7) c) (5,–5) d) (6,–7)
7. The position of a particle moving along a horizontal line of any time t is given by $s(t) = 3t^2 - 2t - 8$. The time at which the particle is at rest is
   a) t = 0 b) $t = \frac{1}{3}$ c) t = 1 d) t = 3
8. Linear approximation for g(x) = cosx at $x = \frac{\pi}{2}$ is
   a) $x+\frac{\pi}{2}$ b) $-x+\frac{\pi}{2}$ c) $x-\frac{\pi}{2}$ d) $-x-\frac{\pi}{2}$
9. The area between $y^2 = 4x$ and its latus rectum is
   a) $\frac{2}{3}$ b) $\frac{4}{3}$ c) $\frac{8}{3}$ d) $\frac{5}{3}$
10. The solution of $\frac{dy}{dx}=2^{y-x}$ is
    a $2^x + 2^y = C$ b) $2^x - 2^y = C$ c) $\frac{1}{2^x}-\frac{1}{2^y}=C$ d) x + y = C

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11. The random variable X has the probability density function $f(x) = \begin{cases} ax+b & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$ and $E(X) = \frac{7}{12}$ then a and b are respectively

a) 1 and $\frac{1}{2}$ b) $\frac{1}{2}$ and 1 c) 2 and 1 d) 1 and 2

12. Which one of the following is a binary operation on N?

a) Subtraction b) Multiplication c) Division d) All the above

13. If in 6 trails, X is a binomial variate which follows the relation 9P(X = 4) = P(X = 2), then the probability of success is

a) 0.125 b) 0.25 c) 0.375 d) 0.75

14. If $v(x,y) = \log(e^x + e^y)$, then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to

a) $e^x + e^y$ b) $\frac{1}{(e^x+e^y)}$ c) 2 d) 1

15. If $\frac{\Gamma(n+2)}{\Gamma(n)} = 90$ then n is

a) 10 b) 5 c) 8 d) 9

16. A is of order n, nonsingular matrix and $\lambda \neq 0$ then adj (λA) =

a) λ^{n-1}adj(A) b) λ^{n-2}adj(A) c) $\frac{1}{\lambda}$adj(A) d) λ^nadj(A)

17. If $\sqrt{-1} = i$, $n \in N$ then

a) $i^{4n+3} = -i$ b) $i^{8n+2} = 1$ c) $i^{100n+4} = -1$ d) $i^{4n+5} = 1$

18. The perpendicular distance from a point with position vector $\vec{u}$ to the plane $\vec{r}.\vec{n} = p$ is

a) $\delta = \frac{|\vec{u}.\vec{n}-p|}{\vec{n}}$ b) $\delta = \frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$ c) $\delta = \frac{|d|}{\sqrt{a^2+b^2+c^2}}$ d) $\delta = \frac{\vec{u}.\vec{n}-p}{|\vec{n}|}$

19. The point on the curve $y = x^2$ is the tangent parallel to X axis is

a) (1,1) b) (1,0) c) (0,1) d) (0,0)

20. Which of the following is not a binary operation on R?

a) + b) – c) ÷ d) x

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

22. Simplify. $i \cdot i^2 \cdot i^3 \ldots i^{2000}$

23. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ be any two boolean matrices of the same type.

Find $A \vee B$ and $A \wedge B$.

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24. Determine whether $x + y - 1 = 0$ is the equation of a diameter of the circle $x^2 + y^2 - 6x + 4y + c = 0$ for all possible values of c.

25. Show that the lines $\frac{x-1}{4} = \frac{z-y}{6} = \frac{z-4}{12}$ and $\frac{x-3}{-2} = \frac{y-3}{3} = \frac{5-z}{6}$ are parallel.

26. Evaluate: $\lim_{x \to 1} \left(\frac{x^2-3x+2}{x^2-4x+3} \right)$

27. Find df for $f(x) = x^2 + 3x$ and evaluate it for $x = 2$ and $dx = 0.1$

28. Show that $y = e^{-x} + mx + n$ is a solution of the differential equation $e^x \left(\frac{d^2y}{dx^2} \right) - 1 = 0$

29. A pair of fair dice is rolled once. Find the probability mass function to get the number of fours.

30. If $4\cos^{-1}x + \sin^{-1}x = \pi$ then find the value of x

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. If adj(A) = $\begin{bmatrix} 2 & -4 & 2 \\ -3 & 12 & -7 \\ -2 & 0 & 2 \end{bmatrix}$, find A.

32. If $|z| = 2$ show that $3 \le |z + 3 + 4i| \le 7$.

33. Discuss the maximum possible number of positive and negative roots of the polynomial equation : $9x^9 - 4x^8 + 4x^7 - 3x^6 + 2x^5 + x^3 + 7x^2 + 7x + 2 = 0$.

34. Find the domain of $f(x) = \sin^{-1}\left(\frac{x^2+1}{2x} \right)$

35. Find the equation of the parabola whose vertex is (5, –2) and focus (2, –2).

36. If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar vectors, show that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$

37. Find the values in the interval $\left(\frac{1}{2}, 2 \right)$ satisfied by the Rolle's theorem for the function

$f(x) = x + \frac{1}{2}, \; x \in \left[\frac{1}{2}, 2 \right]$

38. Use linear approximation to find an approximate value of $\sqrt{9.2}$ without using a calculator.

39. Define an operation * on $\mathbb{Q}$ as follows: $a * b = \left(\frac{a+b}{2} \right)$; $a, b \in \mathbb{Q}$. Examine the closure, commutative, associative properties satisfied by * on $\mathbb{Q}$.

40. Prove that $\int_0^{\pi/2} \log(\tan x)\, dx = 0$

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Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) Solve the following system of linear equations, by Gaussian elimination method: $4x + 3y + 6z = 25$, $x + 5y + 7z = 13$, $2x + 9y + z = 1$.

(OR)

b) Solve the equation $(x - 2)(x - 7)(x - 3)(x + 2) + 19 = 0$.

42. a) Solve the equation $z^3 + 8i = 0$, where $z \in C$

(OR)

b) Using vector method, prove that $\sin(\alpha - \beta) = \sin\alpha \cos\beta - \cos\alpha \sin\beta$

43. a) Prove that $\tan^{-1}x + \tan^{-1}\frac{2x}{1-x^2} = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, $|x| < \frac{1}{\sqrt{3}}$

(OR)

b) At a water fountain, water attains a maximum height of 4m at horizontal distance of 0.5m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75m from the point of origin.

44. a) A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

(OR)

b) If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$

45. a) Find the area of the region bounded between the parabola $x^2 = y$ and the curve $y = |x|$.

(OR)

b) If F is the constant force generated by the motor of an automobile of mass M, its velocity v is given by $M\frac{dv}{dt} = F - kv$, where k is a constant. Express v in terms of t given that v = 0 when t = 0.

46. a) Find the mean and variance of a random variable X, whose probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(OR)

b) Prove that $p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$ using truth table.

47. a) Find the parametric vector, non–parametric vector and Cartesian form of the equations of the plane passing through the three non–collinear points (3,6, –2), (–1, –2,6), and (6,4, –2). *(OR)*

b) Find the local extrema of the function $f(x) = 4x^6 - 6x^4$

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